

Star-Bridge: a topological multidimensional subgraph analysis to detect fraudulent nodes and rings in telecom networks

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Abstract—Fraud mechanisms have evolved from isolated actions performed by single individuals to complex criminal networks. This paper aims to contribute to the identification of potentially relevant nodes in fraud networks with a set of fundamental concepts associated with fraud, notably a group of novel features, Bridging, Influence and Control that tend not only to shape the network, but also to promote the nodes to key positions. Whilst traditional methods for fraud detection rely on trying to identify abnormal patterns, via label propagation, node relation to their alters, odd connections, synchronized activities, and similar authorities, this paper proposes STARBRIDGE, a new linear and scalable, ranked out, parameter free method to identify fraudulent nodes and rings. The paper provides an application to the telecommunications domain where we observe that suspicious fraudulent nodes form a star-bridge-star pattern. Over 75% of nodes involved in fraud denote control, bridging centrality and doubled the influence scores, when compared to non-fraudulent nodes in the same role, stars and bridges are chief among these positions. Influence is an effective metric to identify and rank potential fraudulent nodes outperforming most centrality metrics, demonstrated by a network attack benchmarking the most common ones.

Index Terms—Fraud Type, Fraud Enabler, Fraud Networks, Network Control, Bridging Centrality

I. INTRODUCTION

Every network has a purpose and a goal. The structure and node organization inside the network unveils link distribution, strengths and peculiar features often associated to specific types of networks. Fraud networks, like covert networks try to hide its leaders and be resistant to disruption. On one hand, it protects the most important nodes of the network, not only by avoiding direct exposure, but also possible prosecution if any of these nodes are compromised. Distinct positions in the network drives to different judicial outcomes. Nodes with high degree centrality are more likely to be arrested, on the opposite of nodes with high betweenness centrality [1]. Snake like shapes, where nodes are 9 degrees away from each other, are also described as a strategy that minimizes damage to the network, if one of the nodes is exposed [2]. Typically perpetrators or executors of malicious actions,

have the highest chances of being caught, they are peripheral on the network with limited opportunities to increase their degree. In essence the network limits their tie opportunities protecting the most important nodes [3]. Secrecy and security are strategies taken into consideration in the formation of fraud networks, since they increase resilience to attacks and exposure. Sparse and thin networks with weak ties ensure network survivability, while dense interconnected networks with high group centrality are more exposed and vulnerable [4]. In this aspect, number of ties and level of centrality can indicate the role of the node in the network. Informally, we have the following research problem:

A. Problem Definition

• Given

- Who calls whom network
- Fraud labels for some of the nodes

• Find

- Suspicious fraud nodes and networks
- Patterns of ring associations
- A group of features that characterize fraud nodes and networks

B. Contributions

The contributions of STARBRIDGE are as follow:

- **Scalable**, it scales linearly with the network size
- **Ranked Out**, it provides a ranked score for each node in the network
- **Parameter Free**, requires no parameter tuning
- **Novel Discoveries**, STARBRIDGE led to the identification of nodes with association to fraud networks and network patterns that reveal ring associations.

The remainder of the paper is organized as follow. Section II, provides a brief review of related work. Section III, describes the proposed method, STARBRIDGE. Section IV, experiments with the method and analyses the results obtained and finally Section V concludes the paper.

II. RELATED WORK

Figure 1 was obtained from a Telecom Fraud Management System. It contains the network of all nodes that were marked as suspicious fraud. It is possible to identify an emerging pattern in the network: bridge nodes interconnecting stars, a backbone network that interconnects the major hubs of fraud.

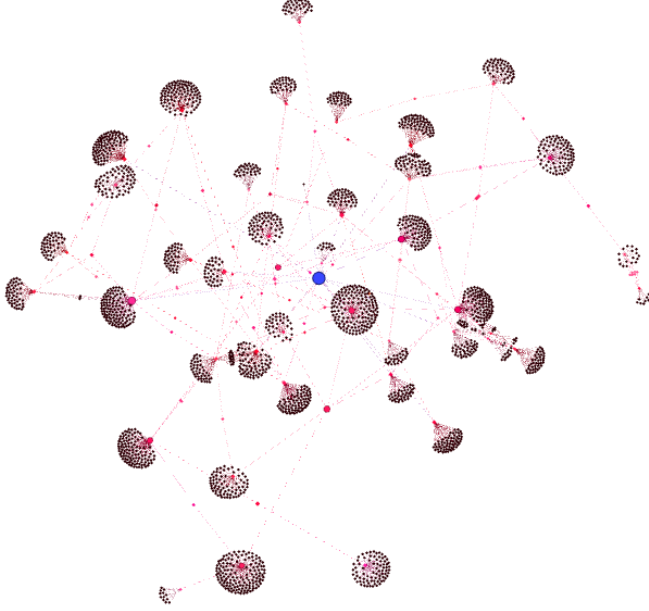


Fig. 1. Suspicious Fraudulent Nodes and Edges [Star-Bridge] pattern.

Previous work [3] highlights how closeness measures the ability of a node to access others in the network, i.e., a supervising role, whilst betweenness demonstrates control over the flow in the network, playing the role of a broker. Furthermore, nodes controlling hubs are usually related to power positions and show the power divisions in the network. Finally, there are nodes that do not have a high degree nor bridge different parts of the network, yet they are strategically well located in the proximity of hubs and large groups, with easy access to other network resource and information. Thus, node centrality, either degree, closeness or betweenness becomes the most common techniques to measure the power of the node in network. However, the suitability of centrality metrics to identify node influence have limitations, namely they fail to value the power of non-hub nodes [5], quantify the difference between them [6] and are highly dependent on the network topology [7]. The accuracy of centrality measures that are optimal to find highly influential nodes, according to their topological properties under a global network context, will most likely be inefficient to capture local information, with respect to the importance of its neighbours where metrics like Eigencentrality and its sub variants (e.g., PageRank, Katz Centrality) are more appropriate. In fact, there are over 200 centrality metrics proposed to date [8] highly dependent in network dynamics and the topological properties driving those dynamics. Furthermore, frequently network centrality mea-

sures are highly correlated, making difficult and insufficient to identify strategic patterns concerning node positioning and role [9]. Regarding network structure, traditionally three different technique classes have been used for modelling structures and communities using graph analysis, namely: **Subgraph Analysis**, **Label Propagation** and **Latent Factor Models**.

A. Subgraph Analysis

Subgraph Analysis is a method that identifies patterns in graphs, with focus on ego-nets and induced subgraphs from its neighbours. They look for “strange nodes” in a social perspective either because they are highly or poorly connected, or reveal “strangeness” in the direction and the weight of its connections, either full inbound links (Blackholes) or full outbound links (Volcanos) or unbalanced weight distribution (heavy links) [10] [11] [12].

B. Label propagation

Label propagation assigns information from labelled nodes to unlabelled nodes through a graph. From an initial subset of labelled nodes, this method constructs a graph based on node-to-node similarity. In the case of nodes connecting to multiple other nodes with different labels, multilevel graphs can be built based on their connected attributes. This class of algorithms aims to identify suspicious nodes based on their connectivity and synchronized behaviour. Real and complex networks tend to have community structures and label propagation methods can find communities by analysing each single entity under the assumption that nodes that are connected in the graph are likely to share the same semantic label. Thus, label propagation algorithms can also infer fraud from legitimate nodes based on their suspicious clique membership and relationships to known fraudsters in the network [13] [14].

C. Latent Factor Models

Finally, Latent Factor Models (LFM's). Latent refers to a group of variables that are inferred from other variables that can be observed. Latent features are computed from observed features using matrix factorization. In a graph context, the observable interactions between a set of nodes can provide information that is useful to predict unobserved links. Latent models are frequently coupled with machine learning and Bayesian methods in order to identify the latent structure associated to the observable variables [15]. From the above, and concerning the dynamics of covert and fraudulent networks, important nodes are a combination of their topological location and local influence, with the intention to not expose the most important nodes, ensuring resilience to network attacks and favourable outcome in case of prosecution. A constant trade-off between risk exposure and ability to control the network [9]. In that realm, STARBRIDGE is a subgraph analysis method, using a multidimensional scale, evaluating each node in their role of Bridge [16], Control [17], and Influence [18] in the network, aiming to rank and identify fraudulent nodes and rings. The first two metrics have dependencies on the topology of the network, while the third one quantifies the

power of the node in accordance to its neighbours. Combined, these attributes score each node according to their role, local influence and centrality, revealing a strategic score on the multiple vectors of node importance.

III. PROPOSED METHOD

Fraud is a composite force driven by technical schemes and human psychology. In this context, nodes with better strategic positioning that allows them to maintain control, avoid exposure and ensure resilience, are key to succeed. An interplay between roles that we propose to be estimated with Bridging Centrality, Control and Influence.

A. Control (Driver Nodes)

To ensure a specific goal in a network the relations between the nodes must support a set of mechanisms or processes. This is often accomplished through controllability, i.e., the ability to steer a system into an arbitrary final state in a finite time [17]. Nodes involved in control, influence the evolution of a system through control strategies, characterized by controllability profiles [19]. Albeit most real systems in terms of processes are nonlinear; in many aspects they are structurally similar and governed by linear and time-invariant dynamics.

$$\frac{dx(t)}{dt} = Ax(t) + Bu(t) \quad (1)$$

The state dynamics of a system of N nodes at a time t , is characterized by its adjacency matrix A , and a set of controlling nodes (or drivers), B , and control signals. A system is controllable if it meets Kalman's controllability rank condition. Hence, equation (1) needs to respect the condition on equation (2) where its controllability matrix, C , has full rank.

$$\text{rank}(C) = \text{rank}(B, AB, A^2B, \dots, A^{(N-1)}B) = N \quad (2)$$

Typically, hubs are nodes that play a chief role in control processes and are typically identified as drivers in matrix B [17]. While Structural Controllability accesses whether control is theoretically possible in a network, nodes with control reveal different properties in distinct graph models. Previous work identified different node level controllability metrics [19], which highlights how certain nodes, may be predisposed to drive the system in a multitude of states, using average, modal, and boundary controllability strategies. While this work was originally performed on brain networks [20], it was further extend to other types of graph models and consequently network structures [19] [21] [22] [23]. One important concept is the Boundary controllability [20], which describes which driver nodes (boundary controllers) can act to synchronize or desynchronize communities in the network. A high number of real world networks have a scale free topology (Barabási-Albert model), and results comparing distinct graph models demonstrated that Barabási-Albert models have higher boundary controllability values and variance [19], suggesting

the correlation between network topology and the role of these nodes in real world networks. Procedures to detect boundary controllers vary from bi-partitioning the original network recursively into smaller subnetworks using Fiedler eigenvector [24] and identify the nodes whose connections span both halves, or by using the modularity function from Newman and Girvan [25]. Due to the intrinsic characteristics of fraud described in section I (trade-off between risk exposure and ability to control the network), we followed the Bridging Centrality metric procedure to identify nodes serving as gatekeepers and which of those are boundary controllers.

B. Bridging Centrality (GateKeeper Nodes)

Nodes with high bridging centrality are very well located in the network. They have a very low degree but serve as bridges between highly connected clusters, functioning as gatekeepers between these different clusters. The bridging centrality metric, for node, i , is obtained from its betweenness centrality, and bridging coefficient [16].

$$C_R(i) = B_C(i).C_B(i) \quad (3)$$

$$B_C(i) = \sum (s, i, t \in V, s \neq i \neq t) \frac{\rho_{st}(i)}{\rho(st)} \quad (4)$$

$$C_B(i) = \frac{d(i)^{-1}}{\sum_{j \in N(i)} d(j)^{-1}} \quad (5)$$

Betweenness centrality (4) defines globally the node importance in terms of the ratio of shortest paths that cross it, (5) bridging coefficient, (3) translates into how well the node is located between high degree nodes, ultimately it gauges the relevance of its neighbourhood, N , via their degree. The product between these two metrics assesses the characteristics of the node in the local network with its relevance in a more global scope. Bridging centrality values are arbitrary since they relate to the network topology, yet, only the top 25 percentile are relevant, since below this point the interest for these nodes rapidly drops [16]

C. Influence (Important Local Nodes)

Node influence metrics quantify the influence of each node within a graph. The most general methods are accessibility, based on the random walks theory, measuring the diversity of self-avoiding walks. Nodes involved in fraud are not necessarily the most influential in terms of their degree or betweenness, yet many fraudulent processes require nodes to reach relevant parts of the network within a limited number of interactions. Due to parametric horizon associated to this metric, not all nodes are equal: only those that are sufficiently close to hubs will score high values. Due to the local nature of this metric, it does not require knowledge of the entire network topology, which may be dynamic, and/or unknown, only local information for each of the nodes is necessary. Fraud networks often create intermediate levels and distance between nodes to conceal fraudulent processes, **influence** analyses these nodes in epidemic-like models over a broad

range of network typologies. Basic SIR Models with close and constant populations can be specified generally by two performance equations, susceptible (6) and infected (7)

$$\frac{dS}{dt} = \frac{-\beta}{N}SI + \gamma I \quad (6)$$

$$\frac{dI}{dt} = \frac{\beta}{N}SI - \gamma I \quad (7)$$

In these equations, β is the average number of contacts made by nodes, N is the total population size and γ is the recovery rate. The Influence Score approach considers an initial scenario of a network where there are no recovered nodes and one seed node i is able to transmit to their clusters in a distance h (typically, a value of 2 is considered for this spreading horizon). In this scenario the neighbours of i , a set with cluster sizes d_1 , contains all first-degree nodes (i.e., those directly infected by i), while d_j is the set of cluster sizes of all infected clusters after j transmissions. As an example, if i contains two neighbours, a and b , the following clusters are formed: after one transmission: $[i \rightarrow a]$, $[i \rightarrow b]$, leading to $d_1 = \{1,1\}$; and after two transmissions $[i \rightarrow a, i \rightarrow b]$, $[i \rightarrow b, i \rightarrow a]$; additionally, if a and b share an edge, the clusters $[i \rightarrow a, a \rightarrow b]$, $[i \rightarrow b, b \rightarrow a]$ are also considered after two transmissions, thus $d_2 = \{2,2,1,1\}$. Assuming non-recovery in the nodes of the network (a fraudulent node does not recover from that state), the Influence Score of a spreading process seeded from node i is a discrete random variable from d_1 to d_h , allowing for the proportionality constant equal to the transmission rate of the process. The Influence Score can be estimated (8) by the entropy of d_j after normalization [18].

$$k(i) = \sum_{j=1}^J d_j \log(d_j) \quad (8)$$

The **Influence Score**, quantifies the distribution of the number of susceptible edges after two transmission events originating from a seed node in a complete susceptible network. The higher the value, the higher the diffusion effect and consequently the influence of the node in the network. Empirically it can be assumed that driver nodes with high bridging centrality have the highest **Influence Score** in the network.

D. Node Role and Influence Labels

Considering the approaches and metrics described in the previous sections, the proposed method labels nodes according to their role in the network. Nodes with control capabilities are labelled with d (driver nodes). Nodes within the top 25 percentile bridging centrality metric are labelled with t (top bridging) and whenever the node reveals both properties a dt label is assigned to the node. Nodes not revealing any of these properties are labelled as n. The Influence Score value is calculated individually to each node. If the dataset as ground proof classification of suspicious abnormal behaviour the label f is added. Consequently, the f label is assigned to nodes with

reported fraud and control; fd, fraud and top bridging; ft and fraud with control and top bridging; fdt. Figure 2 illustrates the applicability of the proposed method.

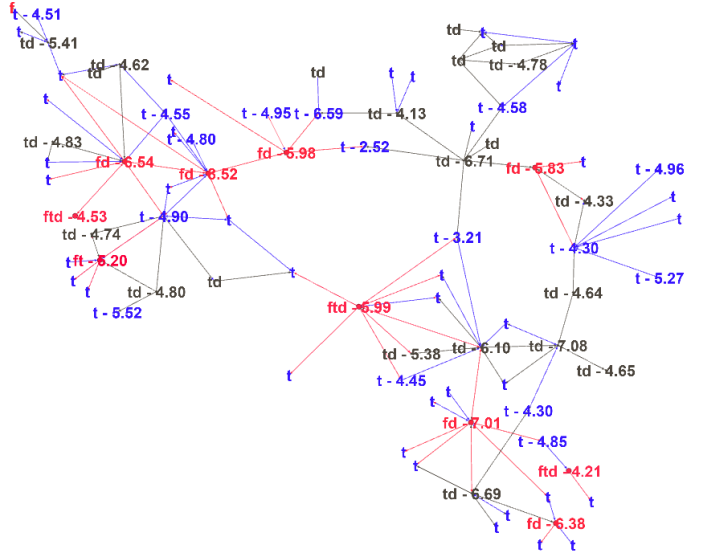


Fig. 2. STARBRIDGE Role Rank.

IV. EXPERIMENT

The dataset used, contains 515.755 nodes and 23.863.734 edges forming 487 connected components. This dataset corresponds to the anonymized who called who network obtained from the records of a telecommunications operator corresponding to all inbound and outbound phone calls during one month billing period. In order to preserve centrality metrics, we focus on the largest connected component in the network, a subset of 28356 nodes and 178542 edges.

In this subset Fraud was identified in 1426 nodes from running an algorithm on a Fraud Management System configured to detect telecom fraud in accordance to TM Forum GB954 [26]. Table I describes the results.

TABLE I
NETWORK NODE ROLE DISTRIBUTION

Complete node set (28356)			Fraud nodes by the FMS(1426)		
Role	%	#	Role	%	#
n	57.34	16262	f	13.77	197
d	18.37	5210	fd	50.27	719
t	16.81	4768	ft	10.69	153
dt	7.46	2116	fdt	25.24	361

Fraud was triggered for nodes with suspicious behaviour, either by abusive usage to international premium numbers, known as (IRSF), International Revenue Share Fraud or suspicious use of a payment method (e.g. a credit card being used by multiple different subscribers or identified as compromised). These types of fraud are known to be performed by organized groups. The influence score was applied to all nodes in the network, as well as the STARBRIDGE dimensions, complemented with the fraud labelling from the Fraud Management

System. Table 1 describes these results. For each different STARBRIDGE category role, or combination of roles, the absolute number and percentage of nodes in each category is provided for the complete node set and for the nodes associated to fraud by the FMS system. It's possible to observe that over 75% of the fraud detected by the Fraud Management System was originated in nodes associated to **control** and **bridging**, highlighting the importance of these roles in fraud. In the violin plot in figure 3, it is possible to compare the influence score of the nodes with and without fraud. While outliers exist on both domains, it is clear to identify a significant gap between the two in terms of influence, showing that fraud nodes have a considerable higher amount of influence than non-fraud nodes.

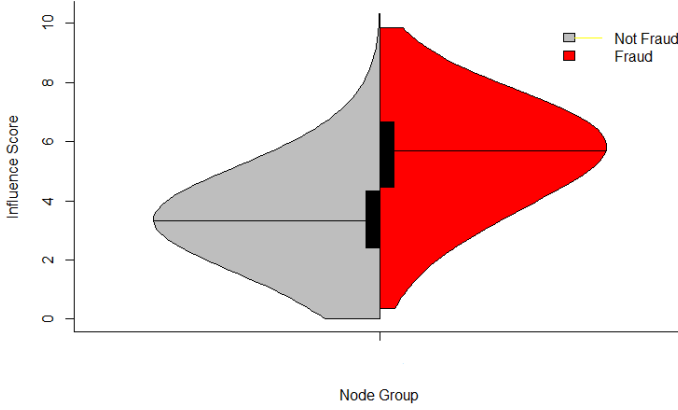


Fig. 3. STARBRIDGE Influence values for Fraud/ Non Fraud Nodes

Figure 4, breaks down Influence by the different roles. It is possible to observe the same gap, notably between control (*d*) and bridging driver roles (*dt*).

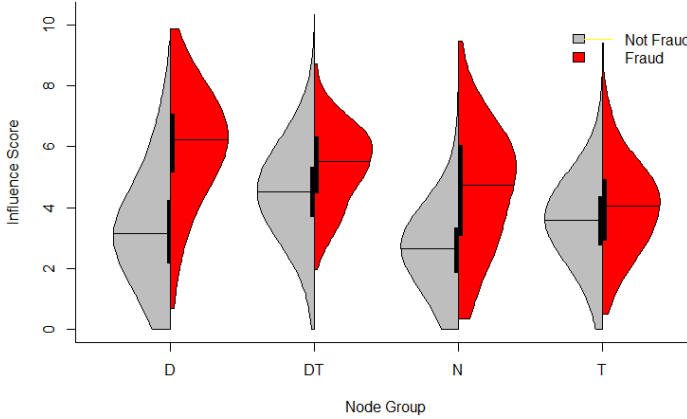


Fig. 4. STARBRIDGE Influence values for Fraud/ Non Fraud Nodes

The value span for Influence is strongly impacted by bridging centrality. This also supports a clear separation of roles, as shown in figure 5 that (*d*) and (*dt*) nodes lead the higher levels of Influence score in the network.

Applying the fraud label to the same set of nodes, it's possible to observe on the scatterplot of figure 5, that the

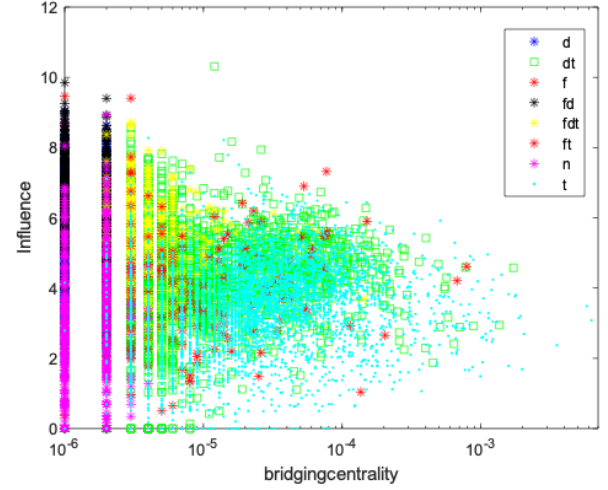


Fig. 5. Influence and Bridging Centrality Correlation (LinLog)

highest majority of fraud was originated from nodes with control role (*fd*) followed by nodes with control and bridging roles (*fdt*).

TABLE II
NODE ROLE INFLUENCE SCORE

Node Role	η .25%	μ	η .75 %
n	1.86	2.65	3.32
d	1.44	2.31	3.15
t	1.88	2.56	3.22
td	2.62	3.94	3.28
Role	η .25 %	μ	η .75 %
f	2.23	4.32	5.87
fd	3.50	4.45	5.37
ft	2.04	2.77	3.53
ftd	3.05	3.92	4.68

Table II provides the average μ , η .25% and η .75% percentile values for the influence score of the nodes with (top) and without fraud (bottom). The difference is visible on how nodes related to fraud have a much higher Influence score in all percentiles. This is especially relevant on the category where most fraud was identified (*fd*), revealing how control and high influence scores are associated to fraud.

To further explore the proposed method, figure 6 provides the correlation of the influence score with most used centrality metrics (LogLog).

It is possible to observe that Influence has a positive correlation with all the metrics, yet, the same is not valid for suspicious fraudulent nodes. The only positive relation of fraudulent behaviour is on the out degree and weighted out degree metrics. Exception for a borderline relation where nodes with medium to high betweenness centrality have an incidence of suspicious fraud nodes. The relationship between these metrics can be observed in figure 8. For visualisation, we used modularity to split the network into different communities, to visualize the role of the different nodes in the network and how they connect. Red nodes represent the suspicious

Correlation between influence and centrality metrics (LogLog)

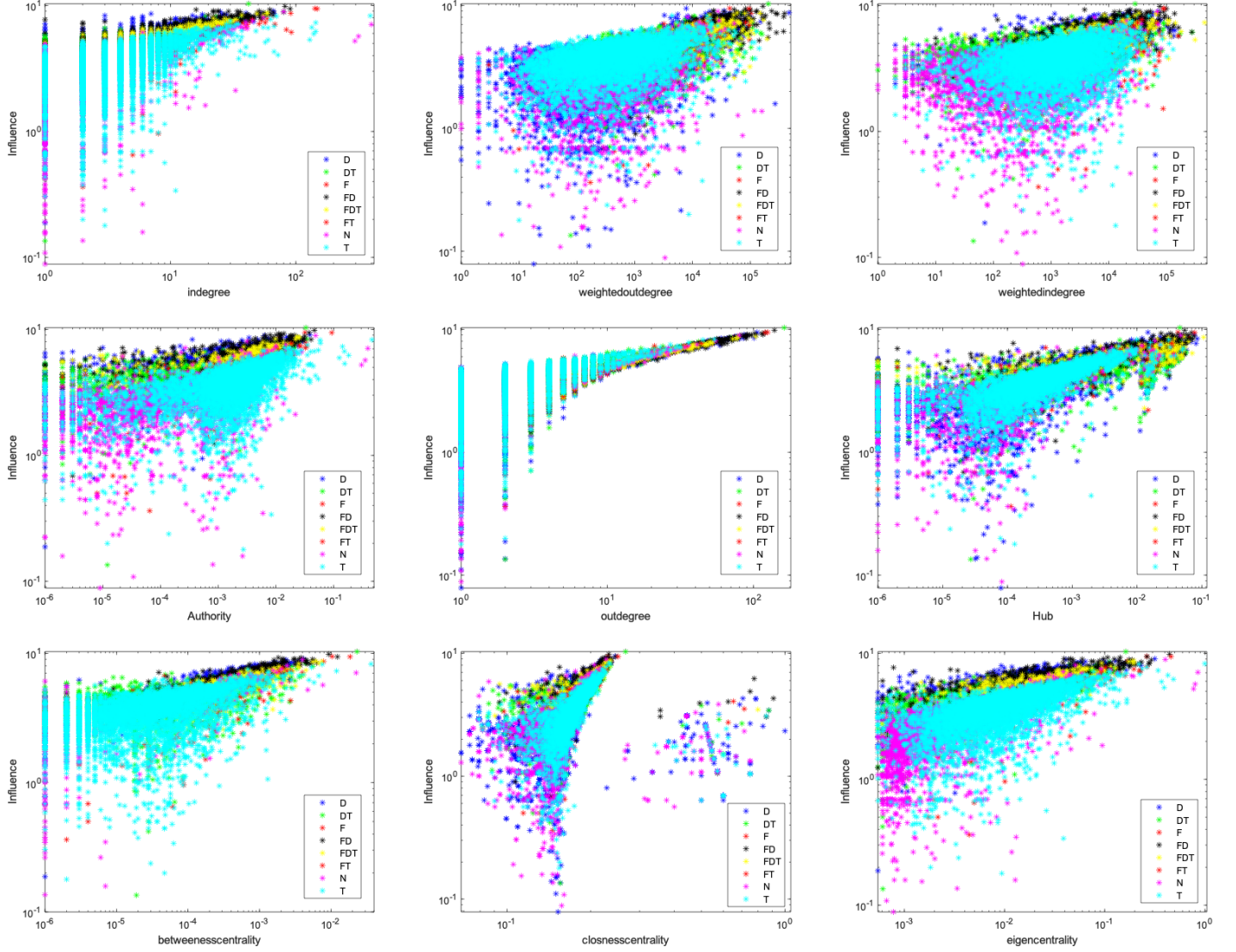


Fig. 6. Correlation between influence and centrality metrics (LogLog)

nodes, (*f,fd,ft,ftd*) while green represents driver nodes (*d*), blue the high bridging centrality nodes (*t*) and black the boundary controllers nodes (*td*). Nodes with no role in the network (*n*) were excluded and are not visible in the community cut.

Suspicious Fraudulent nodes have a higher out degree, forming small hubs, yet, these hubs are interconnected at a distance of 1 or 2 edges by boundary control nodes, with other suspicious fraudulent nodes, which highly increases the influence score (a distance of 2 was used). This is an observation that is consistent in all the 26 communities identified in this dataset. This is complemented by two additional observation also prevalent in all communities that corroborate these values. First, nodes with suspicious fraud (*fd,ftd*), control most of the driver nodes (*d*) and form a path with other fraudulent nodes either directly or using gatekeepers (*t*) and boundary control nodes (*td*). Edges with highest influence (*ft and ftd*) define

the path that connects most of the fraudulent nodes in the network, revealing the bridge and control path used by the suspicious nodes. Out of 1430 suspicious nodes, 958 nodes are connected under the same connected component, spread by 24 communities. These communities tend to be highly connected between themselves, forming a near click, and can only be accessed through specific control nodes, already associated to fraud.

The power of the most influential nodes can be further tested with the robustness of the network. Scale free networks are highly biased structures, in contrast with random based models.

While previous work [27] points betweenness-based strategies, as the most efficient strategy to harm the controllability of real world networks, it is possible to see in table III the simulation of a network attack comparing, Betweenness Centrality, De-

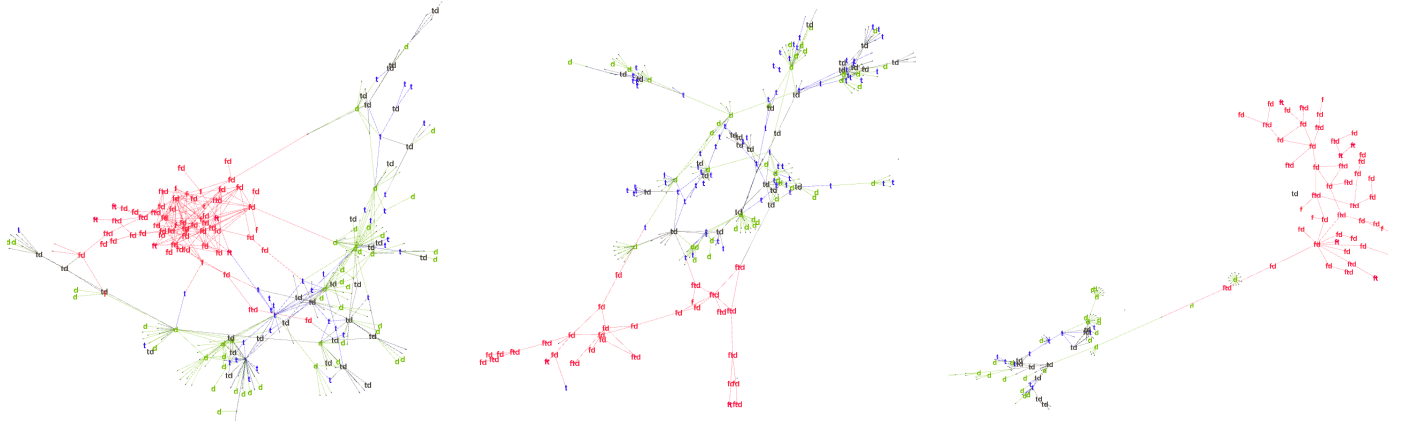


Fig. 7. Nodes in Fraudulent Communities

gree, Influence and Random. The criteria for centrality metrics used relates with benchmarks exposed in figure 7, where out degree and betweenness centrality were the metrics with the highest relation to suspicious fraud classification.

TABLE III
NETWORK ATTACK BENCHMARK (NODE REMOVAL)

Removed%	Betweenness%	Degree%	Influence%	Random%
1	1.53	4.53	1.53	1.53
2	3.33	9.83	3.33	3.33
3	12.99	12.99	6.61	7.55
4	12.99	12.99	6.61	7.55
5	16.10	25.08	54.01	10.76
6	22.41	27.96	57.57	13.9
7	22.41	27.96	57.57	13.9
8	25.08	61.52	70.39	17.01
9	25.08	61.52	70.39	17.01
10	59.6	62.37	71.46	20.02

While nodes with the highest degree and betweenness are the most critical ones for the network collapse until 4%, high influence nodes become the most critical ones after that point. Using a confusion matrix (as represented in figure 8, it is possible to assess the classifier performance of the influence score and the ground truth from the Fraud Management System.

	Actual	
Predicted	True Positive	False Positive
	False Negative	True Negative

Fig. 8. Confusion Matrix

Yet, before looking to the results obtained it is important to highlight the performance reported by Telecom operators [28] regarding Fraud detection using industry solutions.

More than 28% of the telecom operators report a False Positive Rate (FPR) of 90% (Figure 9). In particular, 14% of the operators reported 97% FPR. Concerning True Positive Rate (TPR) 62% of the operators reported a TPR of 80%, nevertheless 7% of the operators, reported a TPR of 60%.

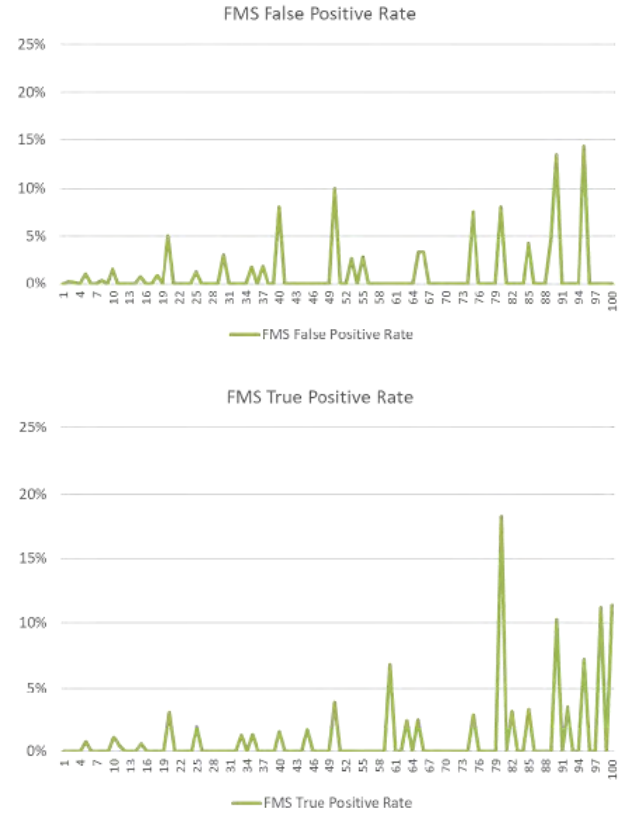


Fig. 9. . FMS False Positive Rate (FPR) and True Positive Rate (TPR), reported by Telecom Operators

While, TPR falls into an acceptable range, even though with some concerning outliers, the same is not true for the FPR. A high number of non fraudulent nodes are considered fraud, implying an untuned or oversensitive system. In order to benchmark the Influence Score with the classification of the Fraud Management System, we used the percentile 75 of table II as the fraud baseline threshold, for the roles (n , d , t , dt).

F	46	78	Accuracy:	0,94	T	116	1195	Accuracy:	0,80
	32	3148	Precision:	0,37		29	4823	Precision:	0,09
			Sensitivity (TPR)	0,59				Sensitivity (TPR)	0,80
			Specificity (TNR)	0,98				Specificity (TNR)	0,80
			Miss Rate (FNR)	0,41				Miss Rate (FNR)	0,20
			Fall Out (FPR)	0,02				Fall Out (FPR)	0,20

FD	553	1088	Accuracy:	0,79	DT	330	1560	Accuracy:	0,64
	100	3990	Precision:	0,34		66	2579	Precision:	0,17
			Sensitivity (TPR)	0,85				Sensitivity (TPR)	0,83
			Specificity (TNR)	0,79				Specificity (TNR)	0,62
			Miss Rate (FNR)	0,15				Miss Rate (FNR)	0,17
			Fall Out (FPR)	0,21				Fall Out (FPR)	0,38

Fig. 10. Influence Score vs FMS Classification Confusing Matrix

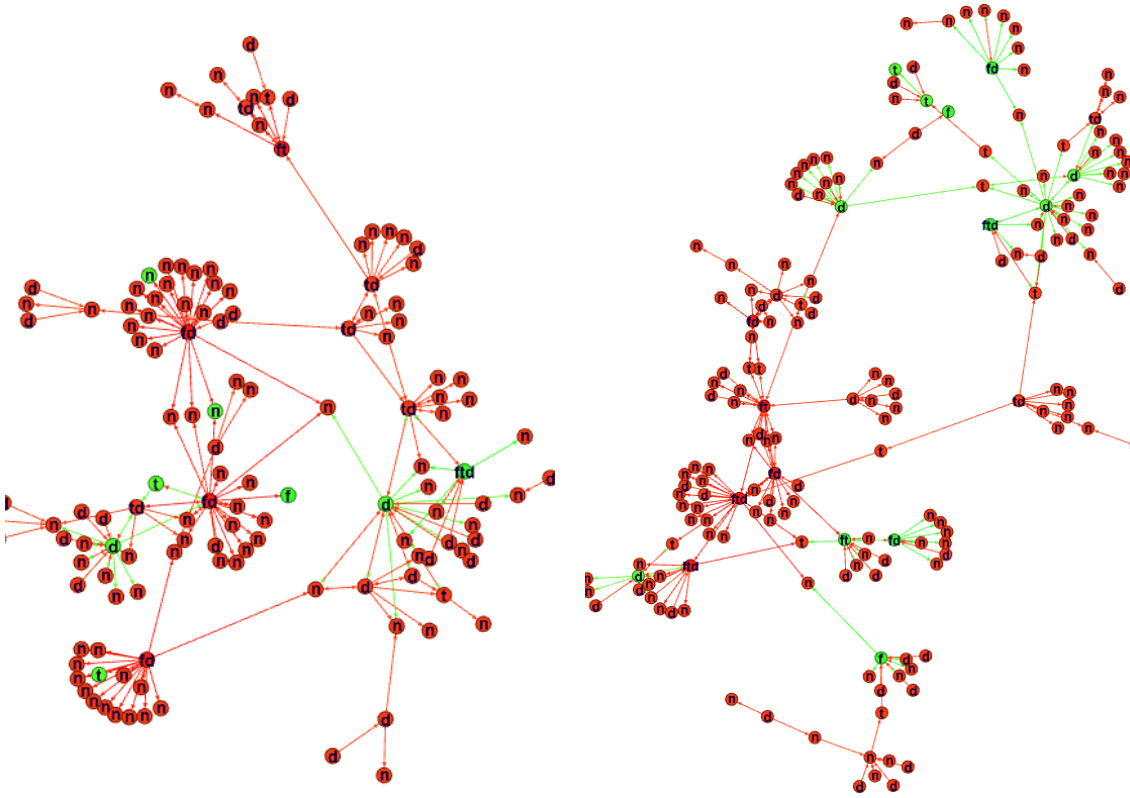


Fig. 11. Modularity cuts classified with TP and TN using STARBRIDGE

On all the different node roles (*d,t,dt*) the TPR rate is above the 80%, in line with FMS industry reported value, yet the FPR is considerably below. While no role in the network (*n*) implies low influence score, the FPR was 2% implying a True Negative Rate of 98%, providing a high reliability. Despite the fact that TPR is 59% and the Miss Rate (FNR) is 41%, it is relevant as future work to identify if these nodes were in fact fraudulent and the fraud management missed, especially in the context that TPR below 80% affects 38% of the operators.

The accuracy and precision values shown in figure 10 for each

role corroborates the symptom that beyond a certain threshold value of influence, the node is associated to suspicious fraud behavior, hence the high accuracy, even though that score values can oscillate in magnitude as described in table 2, therefore a low precision.

Figure 11 illustrates how STARBRIDGE is able to classify correctly each node in the community concerning TP (Green) and TN (Red), based on the percentile 75 of influence for each role

V. CONCLUSIONS

Fraud networks are key factor in the impact of fraud on many industries, namely, in the revenue of telecommunication operators. In order to mitigate the effects of these fraud networks the identification of their members, both ring leaders and partners is chief in many Fraud Management System. Fraudsters need to have key positions that allow them to control and reach other key nodes in the fraud network. These capabilities tend to shape the network based on the roles that fraudulent nodes assume, bridges (t), control (d) and influence. Exploring this concepts it was possible to rank each node in accordance with its neighbourhood and role in the network, revealing its associations but also a consistent pattern on higher influence values when compared to other nodes. This work can be used in conjunction with other abnormal detection algorithms, revealing the most important fraudulent nodes, providing a strategy for node removal with the highest impact, and revealing the associated network and community. It was possible to identify that the combination of these attributes identifies the relevant nodes in the network with the Capacity to perpetrate fraud, not only because of their role in the network but also their influence score. This metric was benchmarked against the classification of a fraud management system on real data revealing that this is in fact a useful metric and that the top 25 percentile of each node group in terms of Influence has a high incidence to become fraudulent. This was complemented by a simulation of a node network attack, where Influence was more effective in network collapsed compared with betweenness centrality. In sum, we can conclude that:

- Influence and controllability are important features.
- Fraud networks tend to have a shape to maximize fraud, we identified star bridge star type of network.
- General and scalable algorithm to spot fraud based in the star-bridge-star (STARBRIDGE).

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